Vishal Kumar Bhagat

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LinkedIn | GitHub | LeetCode | Codeforces

Summary

Full-Stack & Applications Engineer experienced in building scalable web platforms, **RESTful APIs**, and automated workflows. Proficient in **C++, Python, TypeScript, and SQL** with strong foundations in **OOP, Data Structures & Algorithms**, and relational databases. Skilled in designing production-oriented full-stack systems and integrating modern backend services.

Professional Experience

AI Engineering Intern – Meraki Design Agency June – July
Untold India Experiences *Delhi / Remote*

- Built an embeddable **AI travel chatbot** using React, FastAPI, and LangGraph for conversational trip planning and personalized experience discovery.
- Architected a **RAG pipeline with PostgreSQL + pgvector** for context-aware responses, structured trip extraction, and personalized itinerary generation.
- Optimized chatbot performance using **SSE streaming and LRU caching**, reducing repeated-query latency from **450ms to 20ms (96%)** while enabling incremental responses.

Technical Team Member – Hash Define 2024 – Present
College Technical Society *New Delhi, India*

- Taught **30+ students** through technical sessions on Data Structures, Algorithms, and programming fundamentals.
- Organized and executed **technical events**, including speaker sessions, workshops, and coding sessions.
- Organized a **Codeathon**, handling problem-setting, event coordination, and on-ground execution.

Projects

Sentra AI – Autonomous Software Engineering Agent | *GitHub* *React, Node.js, FastAPI, LangGraph*

- Built an autonomous multi-agent code repair platform using **Python, LangGraph, and FastAPI**, converting GitHub issues into tested, human-reviewed Draft Pull Requests.
- Engineered an **AST-driven context skeletonizer**, reducing LLM prompt overhead by **75–85%** and improving Time-to-First-Token (TTFT) by **77%**.
- Developed a **3-tier fuzzy patch engine** using Myers alignment and Levenshtein similarity, increasing patch success from **68% to 96%** and reducing verification to **<0.5s**.

Quant Pilot – AI Financial Analysis Platform | *GitHub* *React, FastAPI, LangGraph, PostgreSQL*

- Architected a stateful **multi-agent financial analysis system** with LangGraph and FastAPI for real-time stock market reasoning.
- Built a trading terminal integrating the **Alpaca Trading API** for live portfolio synchronization and one-click BUY/SELL execution.
- Implemented persistent AI memory and confidence-scored BUY/SELL inference using PostgreSQL, covering **5,000+ ticker symbols**.

DevChat – Real-Time Collaborative Chat Platform | *GitHub* *React, Node.js, WebSockets, Redis, MongoDB*

- Built a real-time chat platform using **Node.js, WebSockets, and React**, supporting persistent one-to-one messaging with low-latency bidirectional communication.
- Implemented **Redis-backed WebSocket infrastructure** for connection management, message delivery, and real-time client synchronization.
- Designed a production-ready backend with **MongoDB persistence, authentication, and containerized deployment** for reliable storage and scaling.

Education

Guru Gobind Singh Indraprastha University (GGSIPU) Expected 2028
B.Tech. in Computer Science Engineering ***– CGPA: 8.6/10*** *Delhi*

Modern School, Vaishali 2024
Senior Secondary Education *Delhi*

Achievements

- Solved **500+ DSA problems** across LeetCode and Codeforces; completed an Advance AI Agent certification by **IBM**
- Maxrating **1091 on Codeforces** and **1500 on LeetCode**.

Technical Skills

Languages: Java, Python, TypeScript, C++, JavaScript, SQL
Frontend: React.js, HTML5, CSS3, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, LangGraph, REST APIs, WebSockets
Databases: PostgreSQL, MongoDB, MySQL, Redis
Tools: Git, GitHub, Docker, Postman, Azure